

INGUINAL CANAL

Department of Anatomy

The abdomen (2)

Second Part

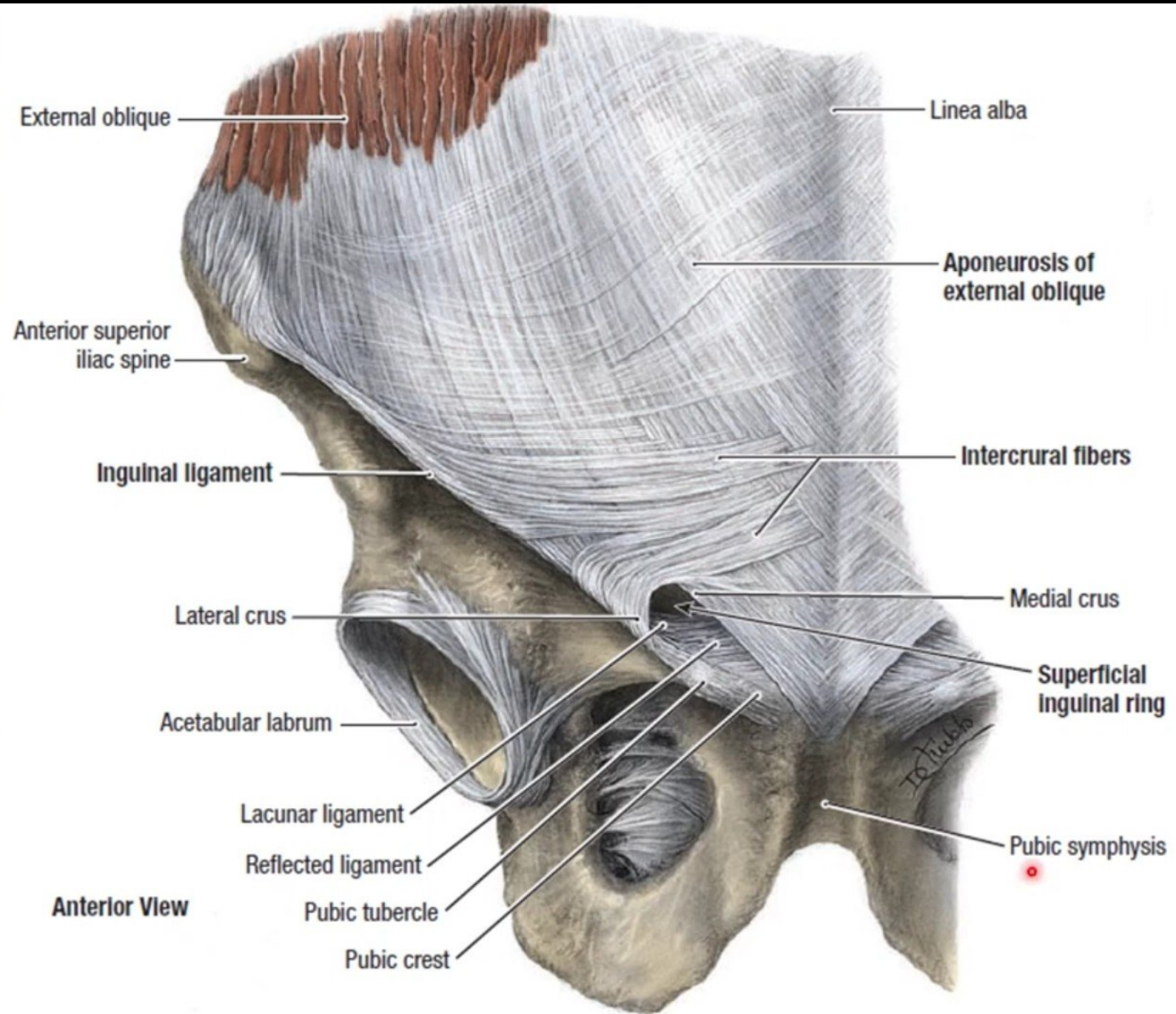
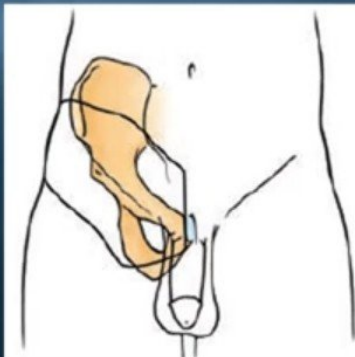
Dr. Osman Ahmed Osman

Lecture points

- Anterior abdominal wall have 8th layers (remember).
- Definition of Inguinal canal.
- Inguinal canal orientation.
- Inguinal ligament.
- Inguinal canal like (a Box !!).
- Deep inguinal ring (Inlet of inguinal canal)
- Superficial inguinal ring (Outlet of inguinal canal)
- The anterior and posterior wall of the canal.
- The roof and floor of the inguinal canal.
- The integrity of the inguinal canal depends upon:
- The Contents of inguinal canal.
- The round ligament of the uterus in females in the inguinal canal
- The ilioinguinal nerve.
- The Main functions of Inguinal canals.
- **The Defensive mechanism (shutter mechanism).**
- Anatomical dissection of inguinal region in male
- Anatomical dissection of inguinal region in female.
- The Spermatic Cord.
- The layers coverings of the spermatic cord
- The contents of the cord consist .
- Descent of gonads in male .
- Descent of gonads in female.
- Applied clinical anatomy.
- Indirect (oblique hernias)
- Direct hernias.
- Pressures in the inguinal canal
- Inguinal triangle.
- Relation of the types of hernias to inguinal triangle
- Summary of clinical correlation
- testicular venogram.

Anatomical dissection of inguinal region in male

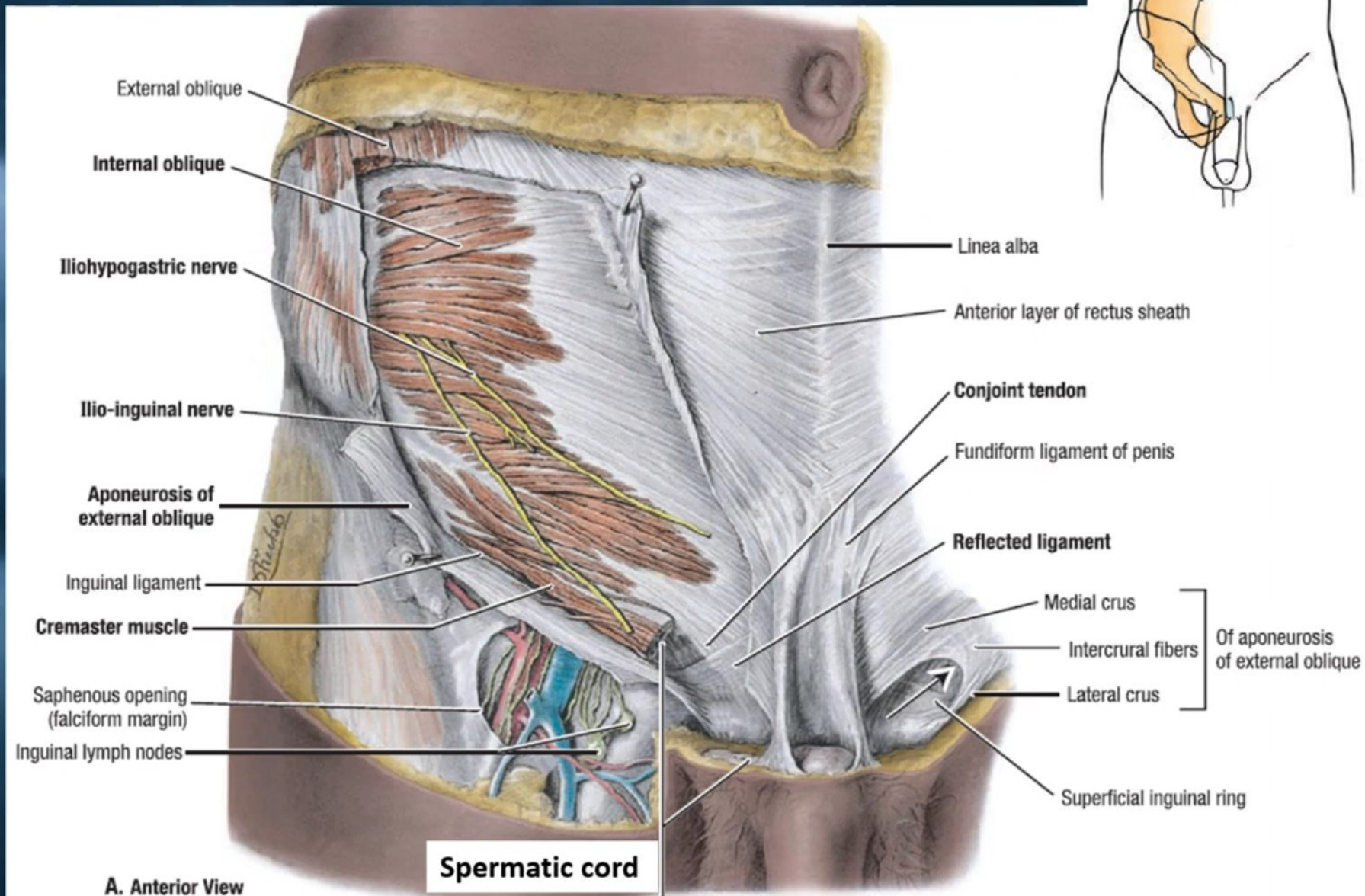
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Anterior View

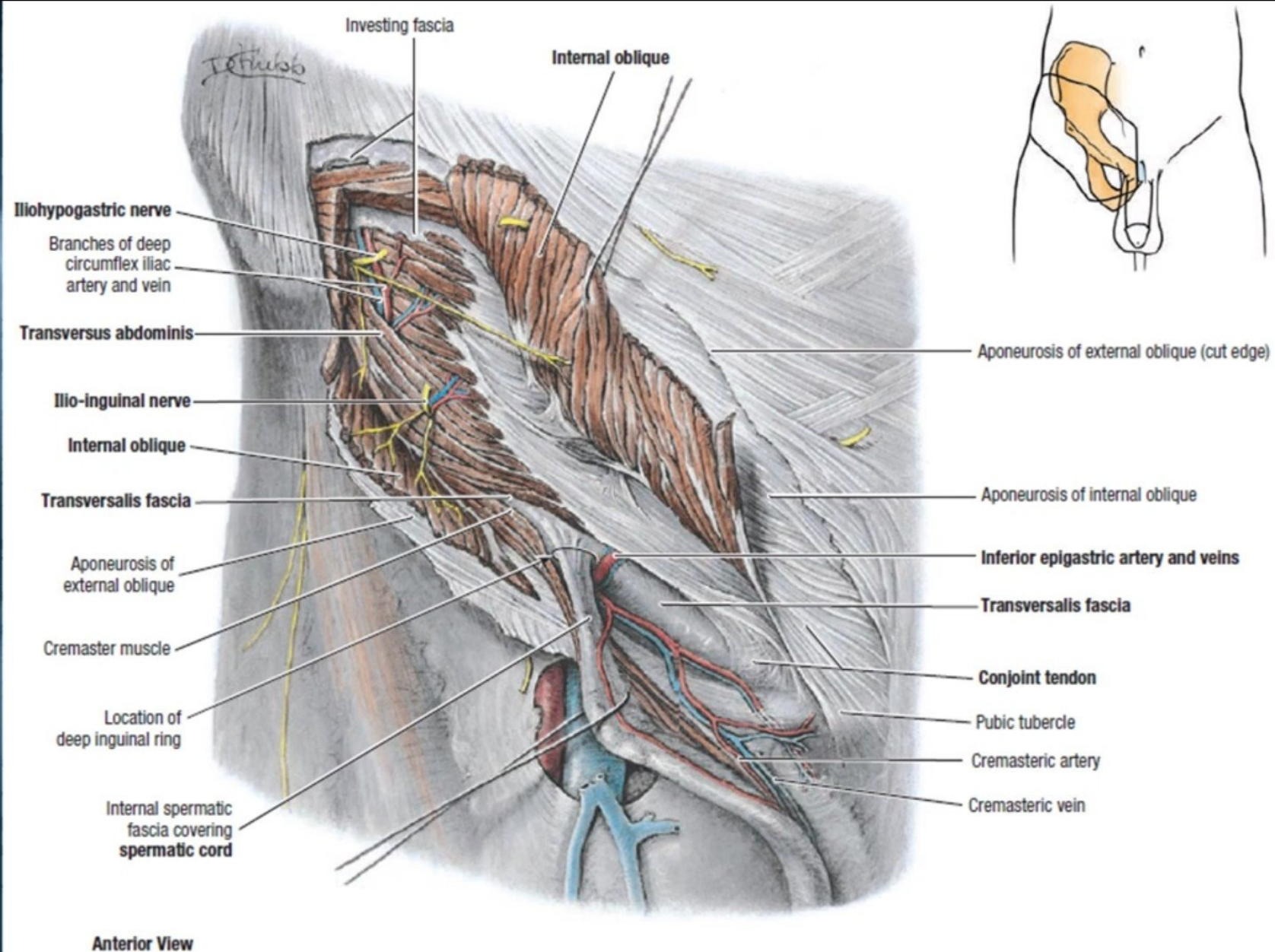
Anatomical dissection of inguinal region in male

(2)



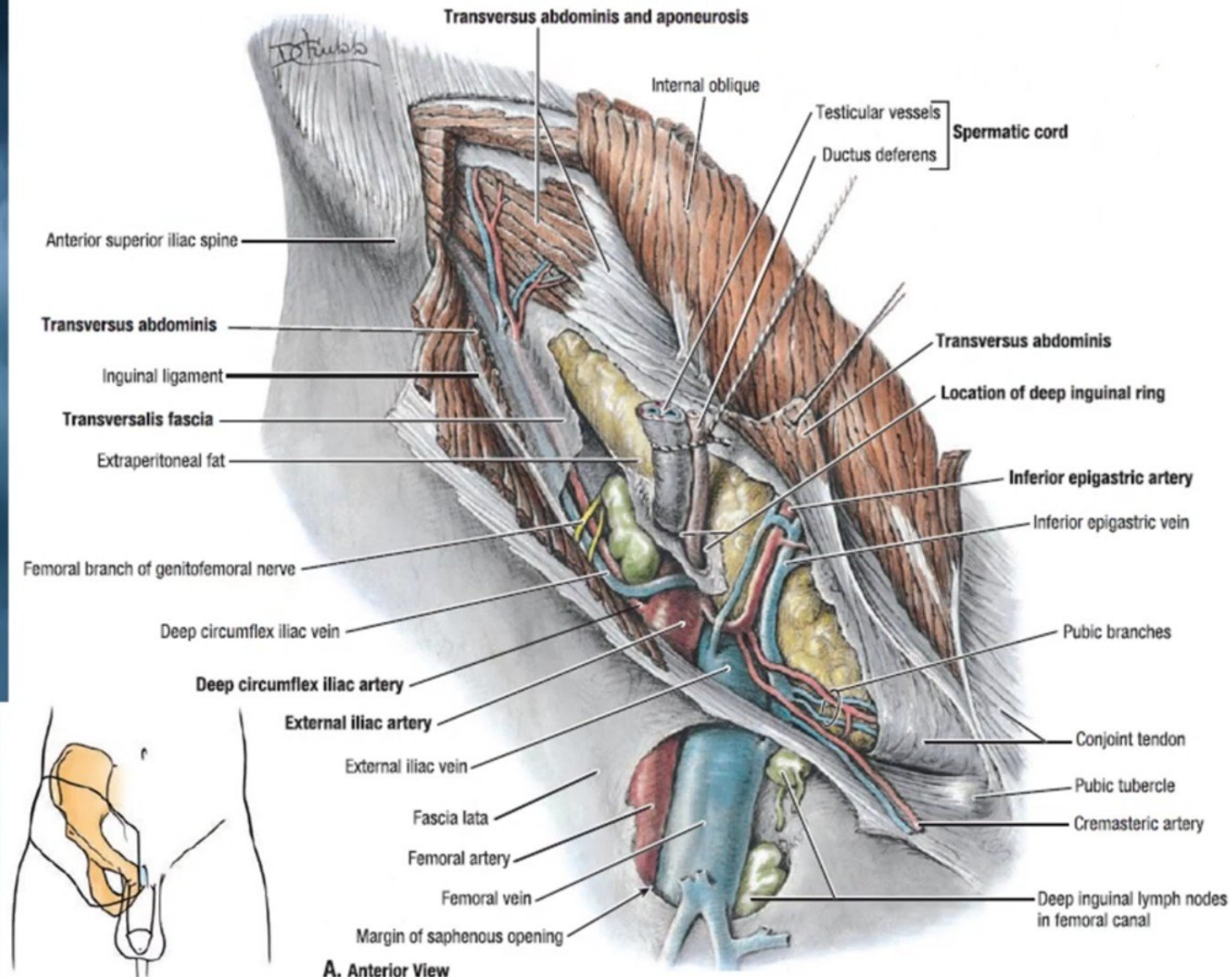
Anatomical dissection of inguinal region in male

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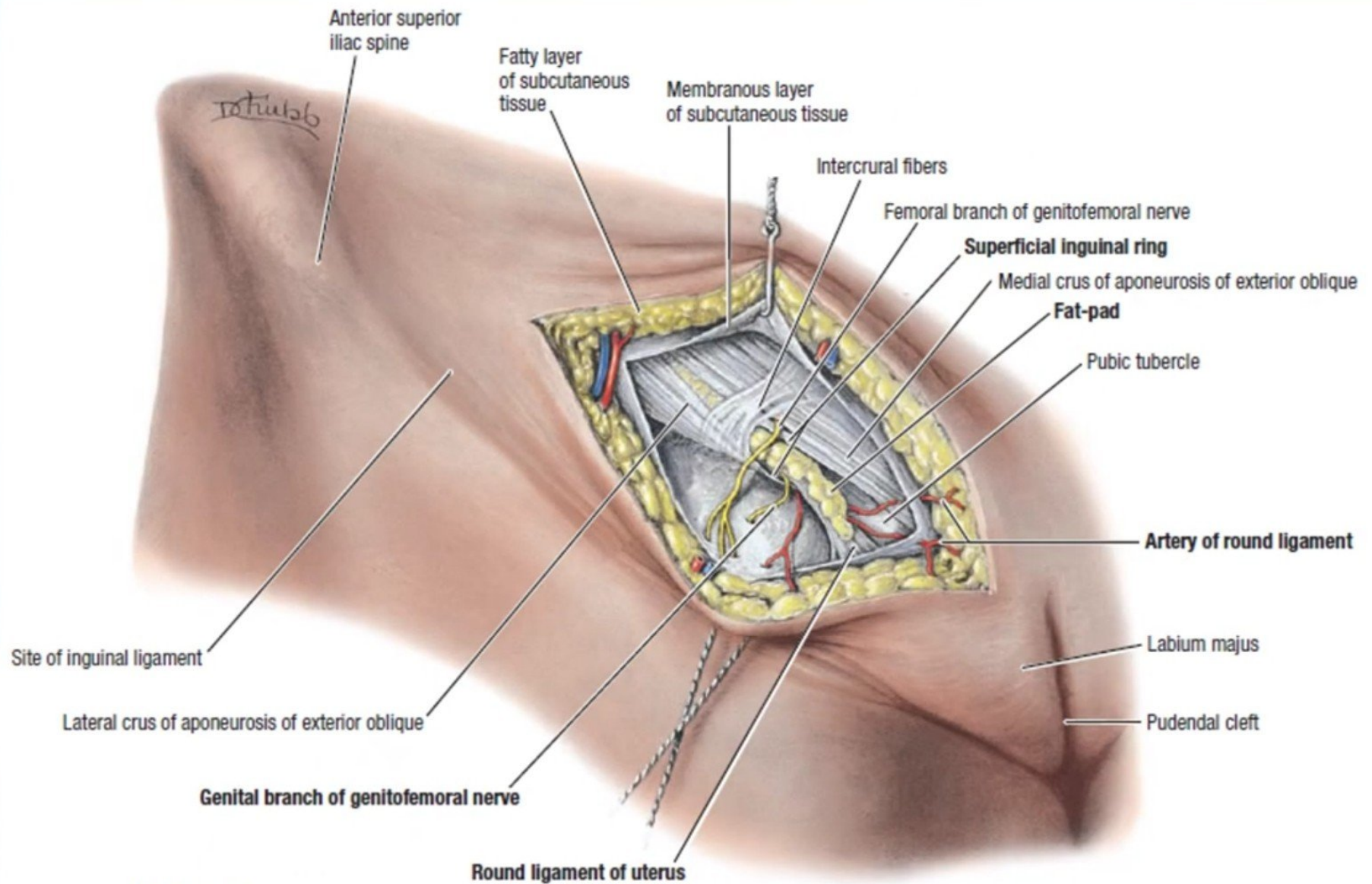
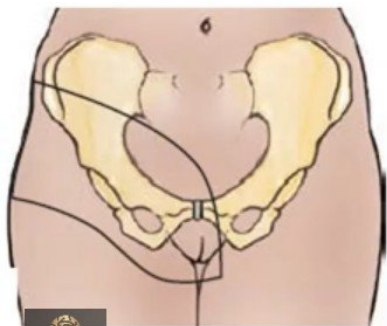
Anatomical dissection of inguinal region in male

(4)



Anatomical dissection of inguinal region in female

(1)



A. Anterior View

Anatomical dissection of inguinal region in female

(2)

Membranous layer of subcutaneous tissue

Internal oblique

Cremaster muscle

Inguinal ligament

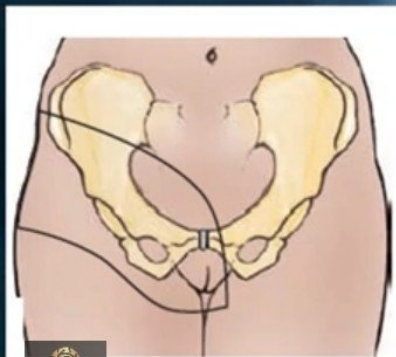
Aponeurosis of external oblique

Femoral branch of genitofemoral nerve

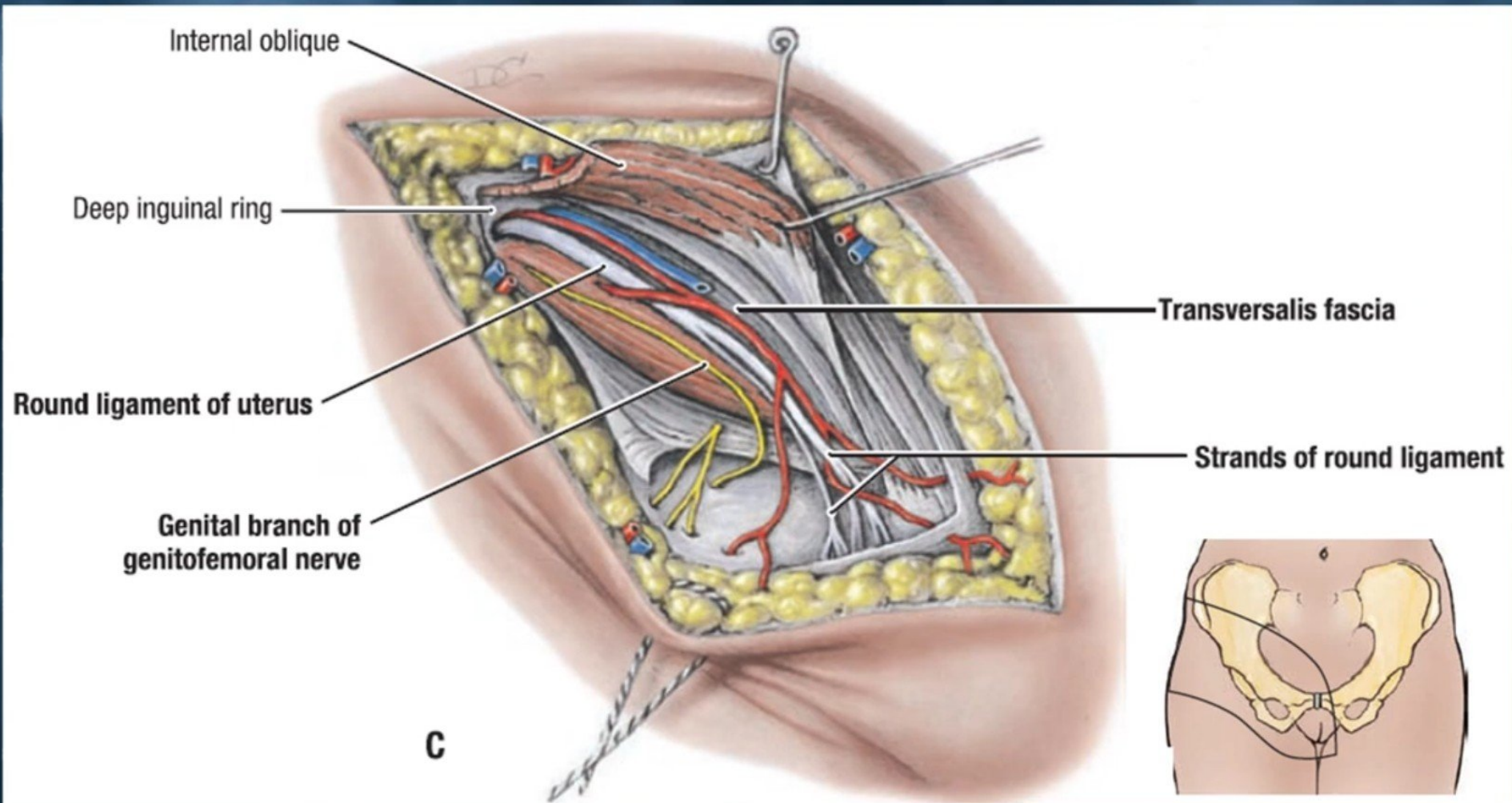
Fat-pad

Strands of round ligament of uterus

B



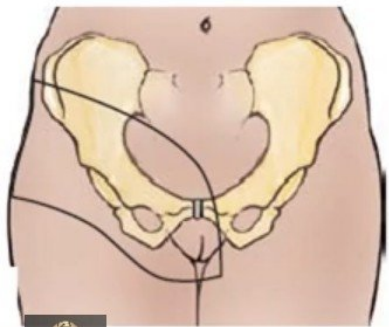
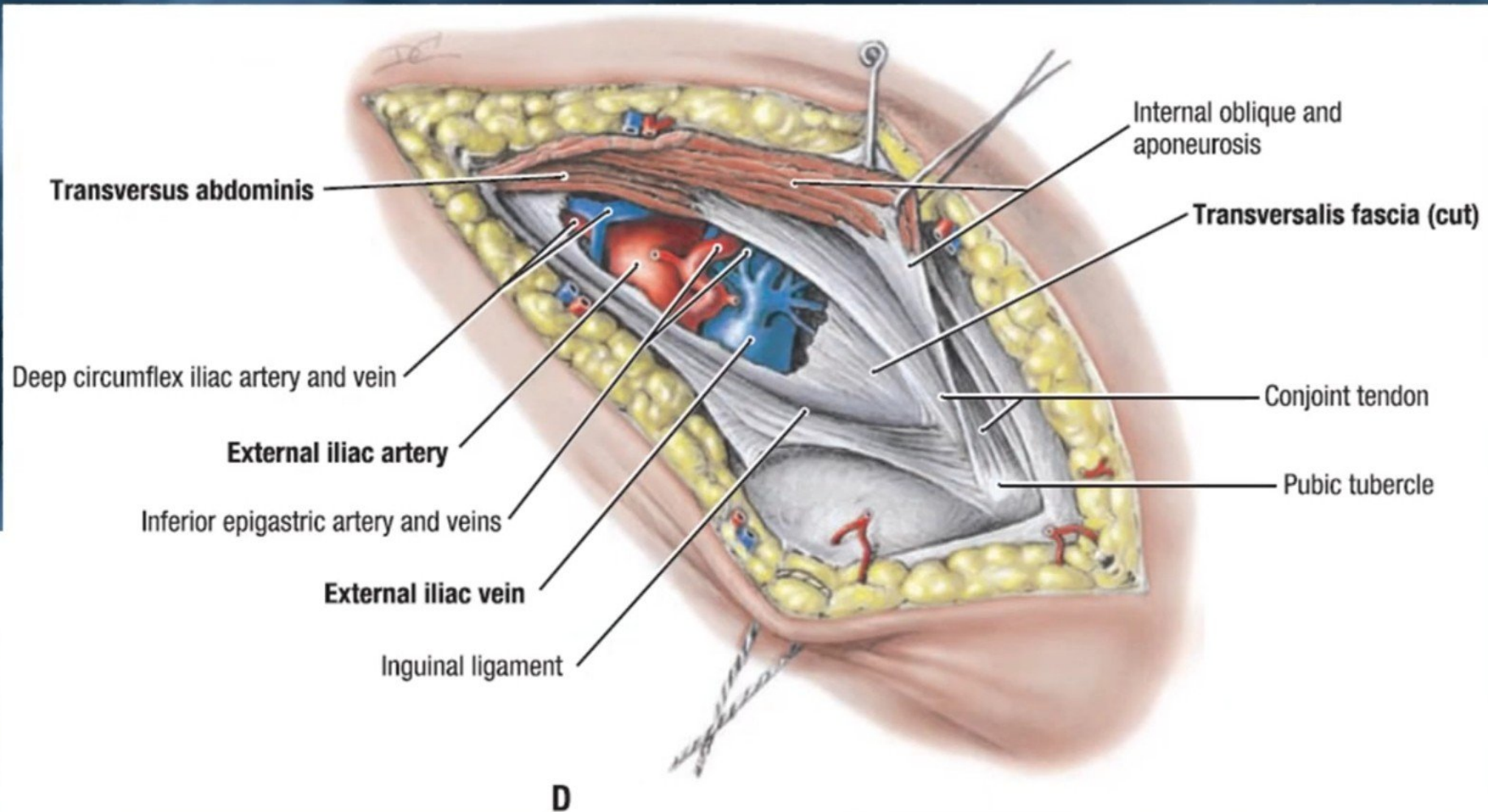
Anatomical dissection of inguinal region in female



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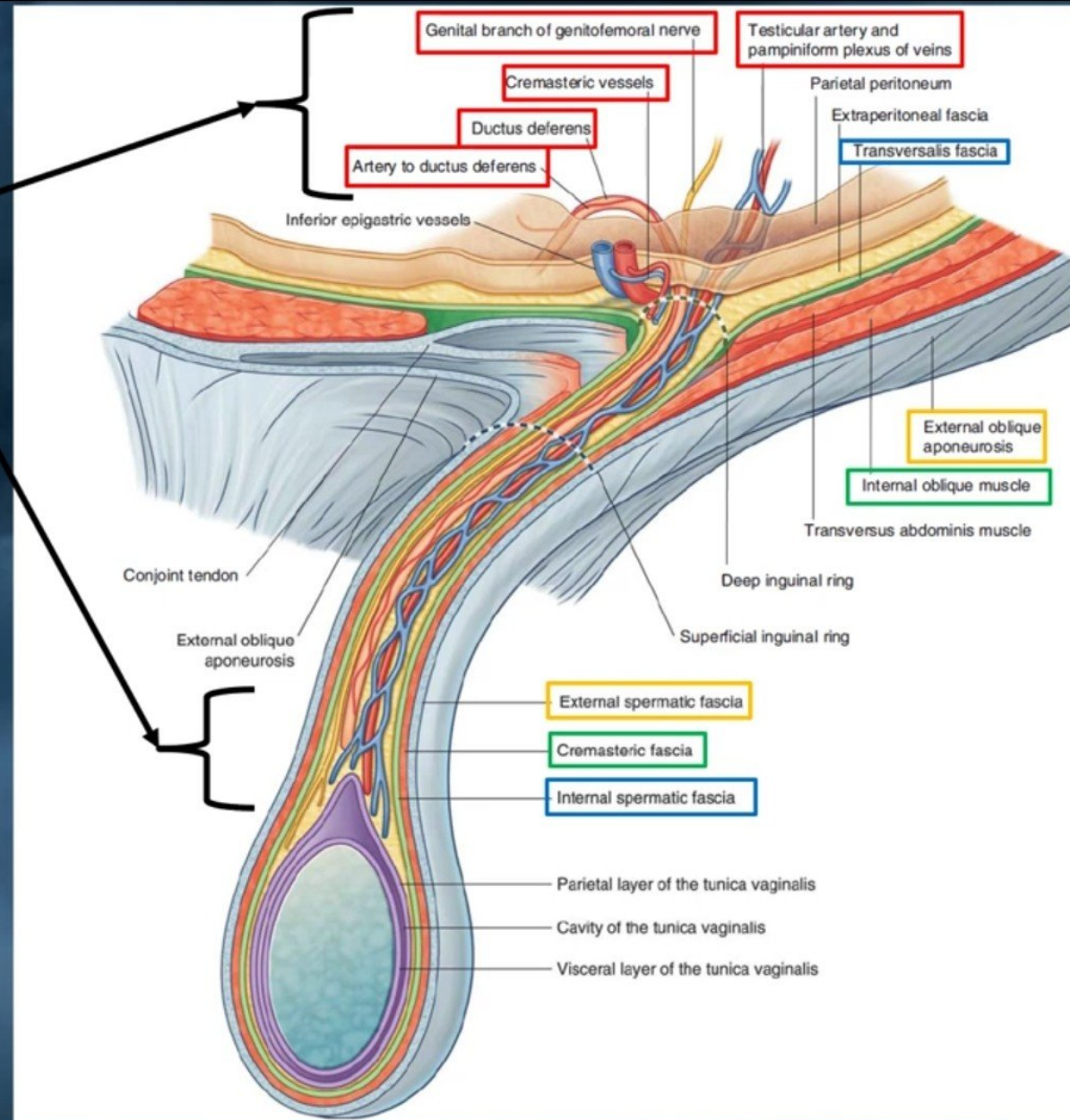
Anatomical dissection of inguinal region in female

(4)



The Spermatic Cord

- The spermatic cord has:
 - Six (groups of) constituents.
 - Three coverings.
- It is a collection of structures that **pass through the inguinal canal to and from the testis**.
- It is **covered with three concentric layers of fascia derived from the layers of anterior abdominal wall**.
- It **begins at the deep inguinal ring lateral to the inferior epigastric artery** and ends at the testis.



The layers coverings of the spermatic cord

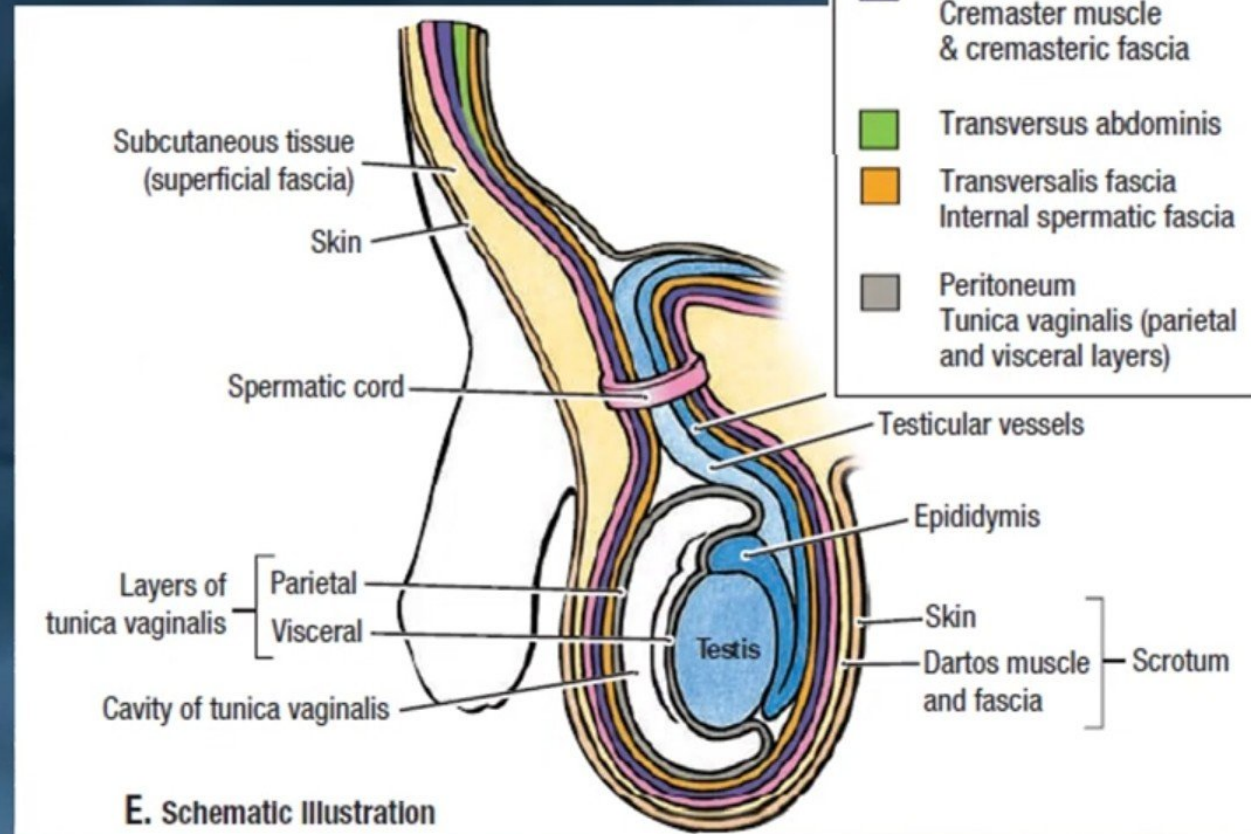
The three coverings of the spermatic cord:

1) The internal spermatic fascia:

- ✓ It is derived from the transversalis fascia at the deep inguinal ring.

2) The cremaster muscle and cremasteric fascia:

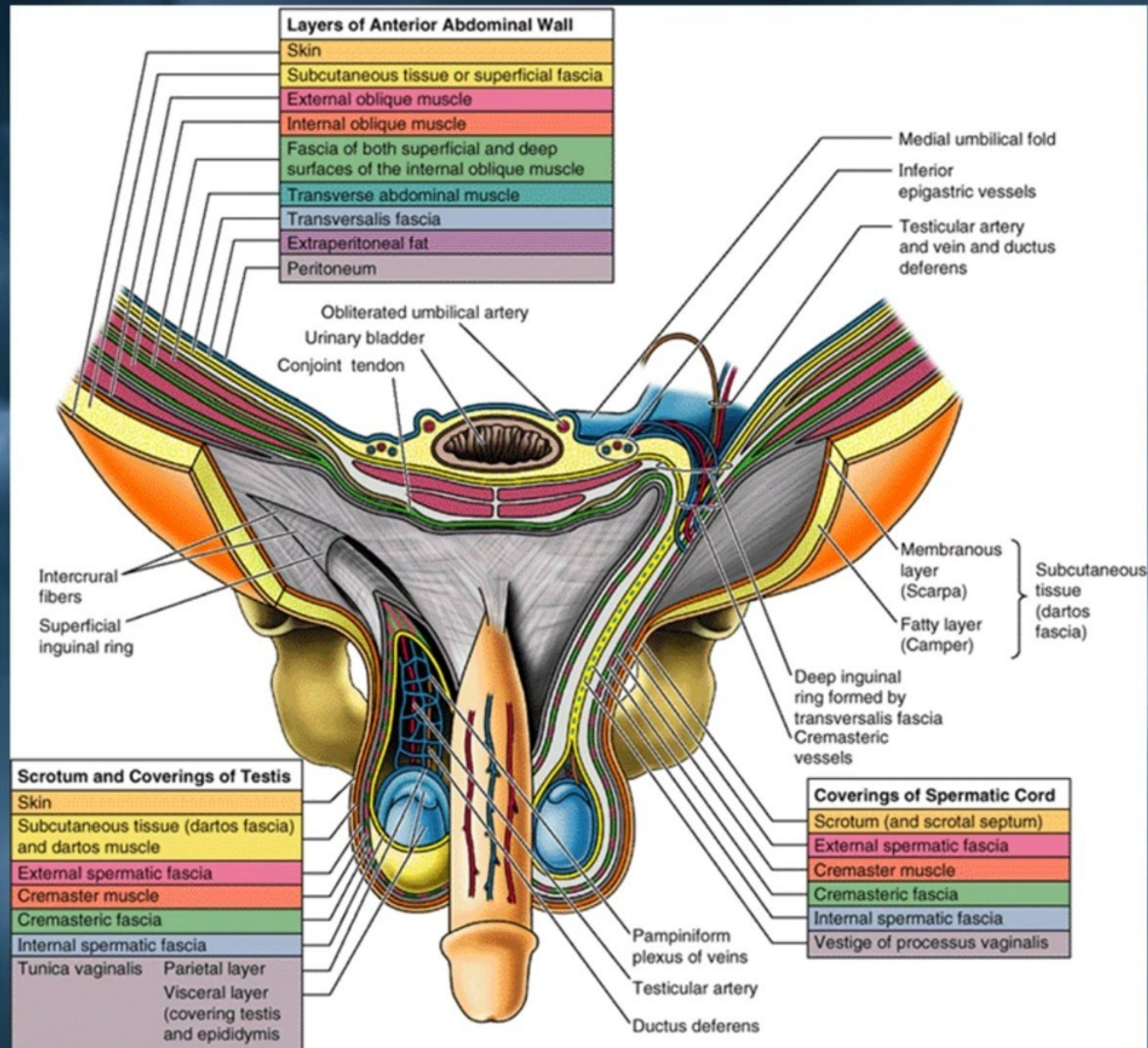
- ✓ This loosely arranged layer consists of striated muscle bundles united by areolar tissue.
- ✓ The muscle arises laterally from:
 - The inguinal ligament.
 - The internal oblique muscle.
 - The transversus abdominis muscle.
- ✓ The fibers spiral down the cord (the longest reaching as far as the **tunica vaginalis** of the testis) and loop back to become attached to the pubic tubercle.



3) The third covering, the external spermatic fascia:

- ✓ It is acquired from the external oblique aponeurosis as the cord passes between the crura of the superficial ring.

The layers coverings of the spermatic cord



The contents of the cord consist of:

1) The vas deferens:

- ✓ It lies in the lower and posterior part of the cord.

2) The artery to the vas artery:

- ✓ It is from the superior or inferior vesical artery.

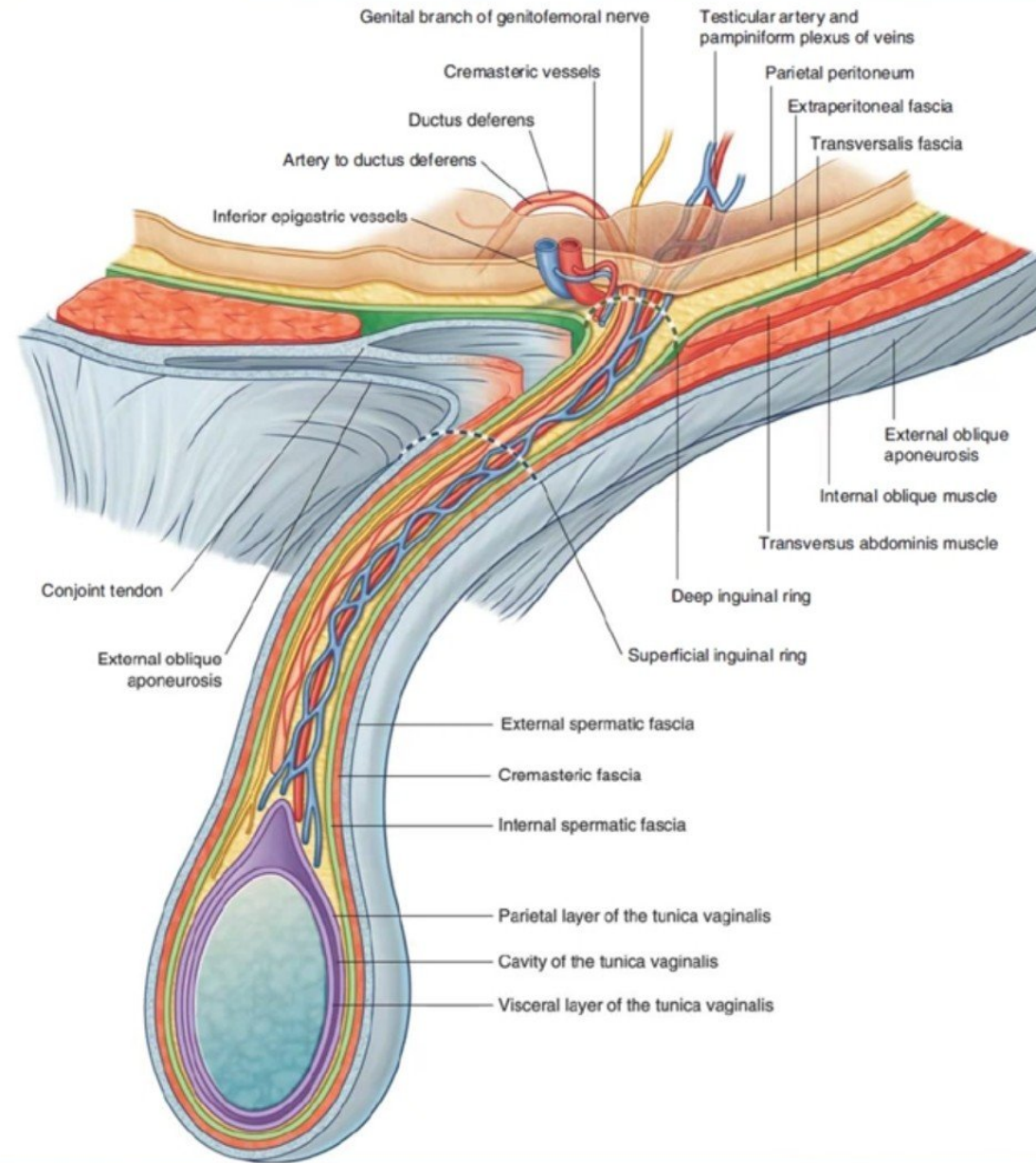
3) The testicular artery:

- ✓ The largest artery from the abdominal aorta.

4) The cremasteric artery:

- ✓ It from the inferior Epigastric. to the coverings.

5) The venous pampiniform plexus



The contents of the cord consist of:

6) The Lymphatics.

- ✓ It is from the testis draining to para-aortic nodes.
- ✓ It is drain to external iliac nodes.

7) The genital branch of the genitofemoral nerve:

- ✓ It supplies the cremaster muscle.

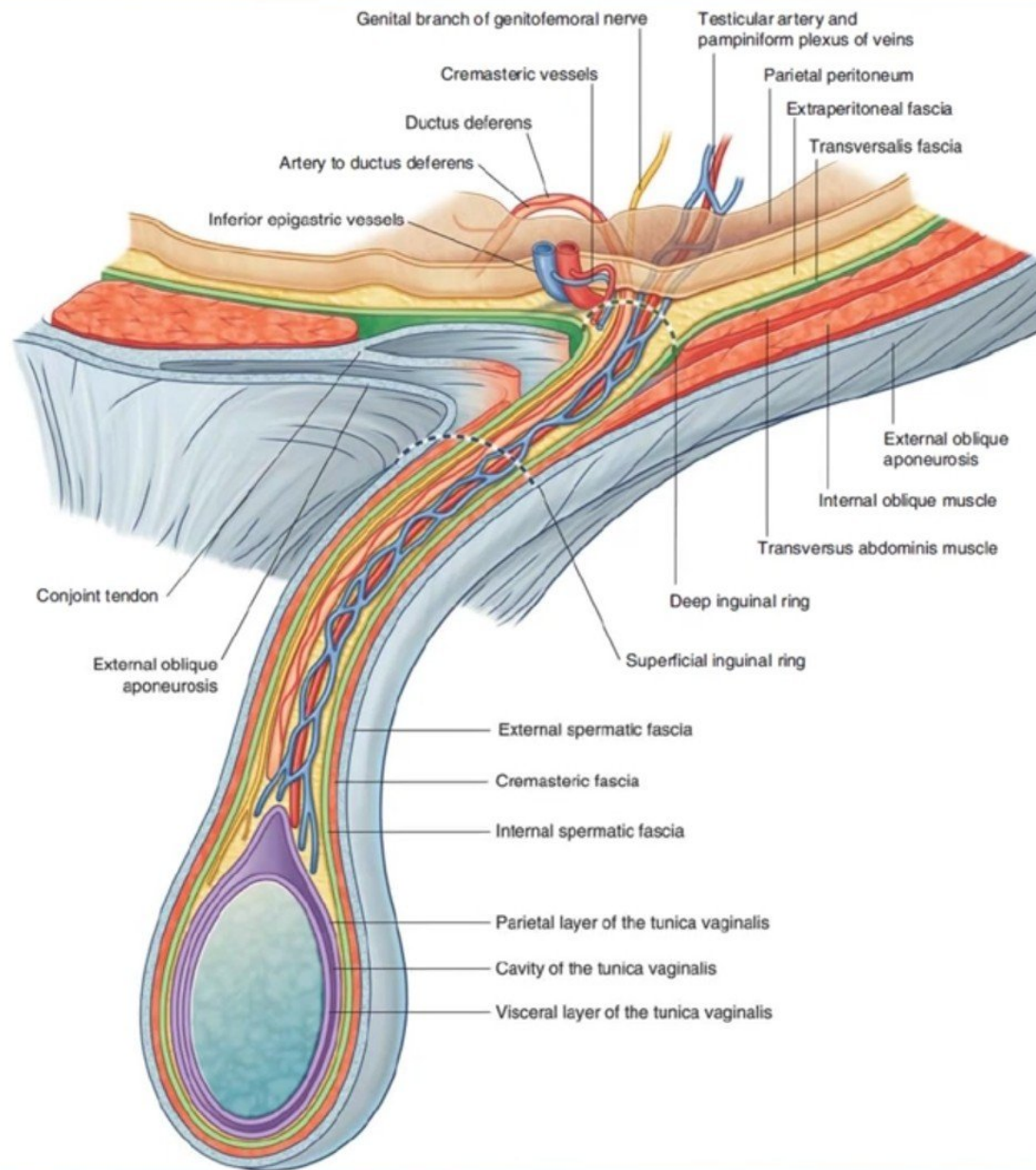
8) The sympathetic nerves:

- ✓ Twigs which accompany the arteries.

The processus vaginalis:

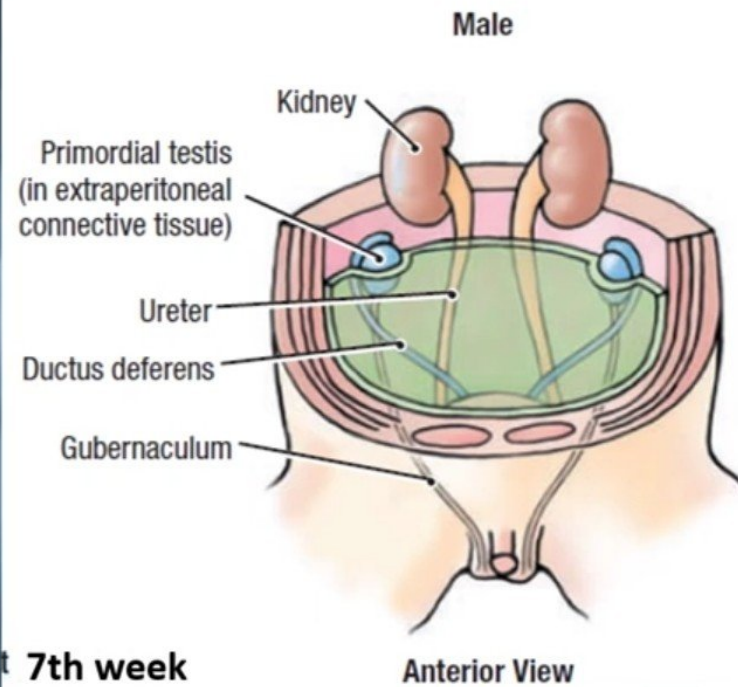
The obliterated remains of the peritoneal connection with the tunica vaginalis of the testis.

(When patent it forms the sac of an indirect inguinal hernia).

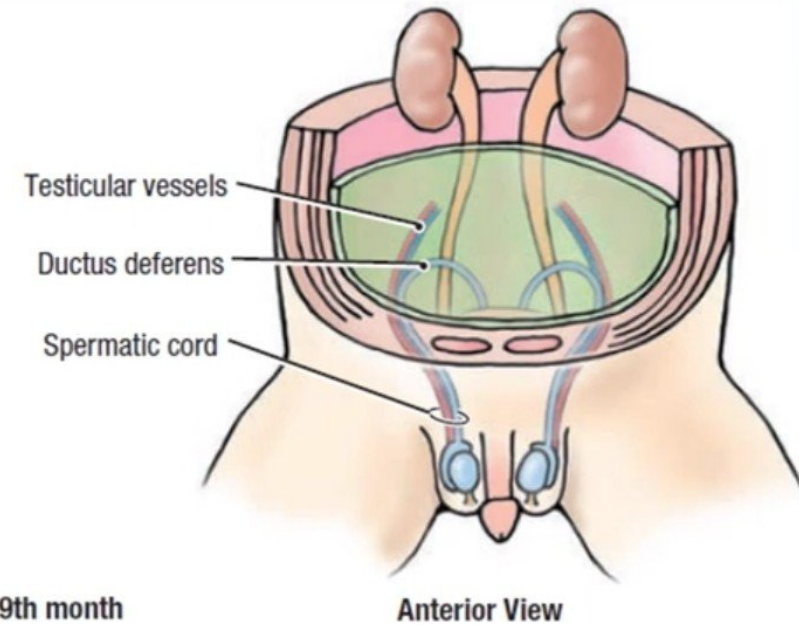


Descent of gonads in male

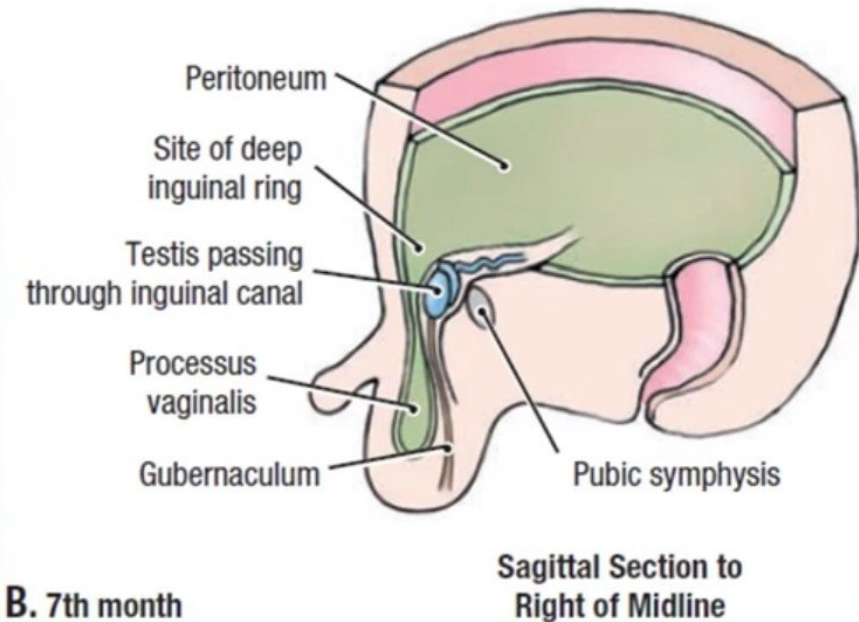
at 7th week



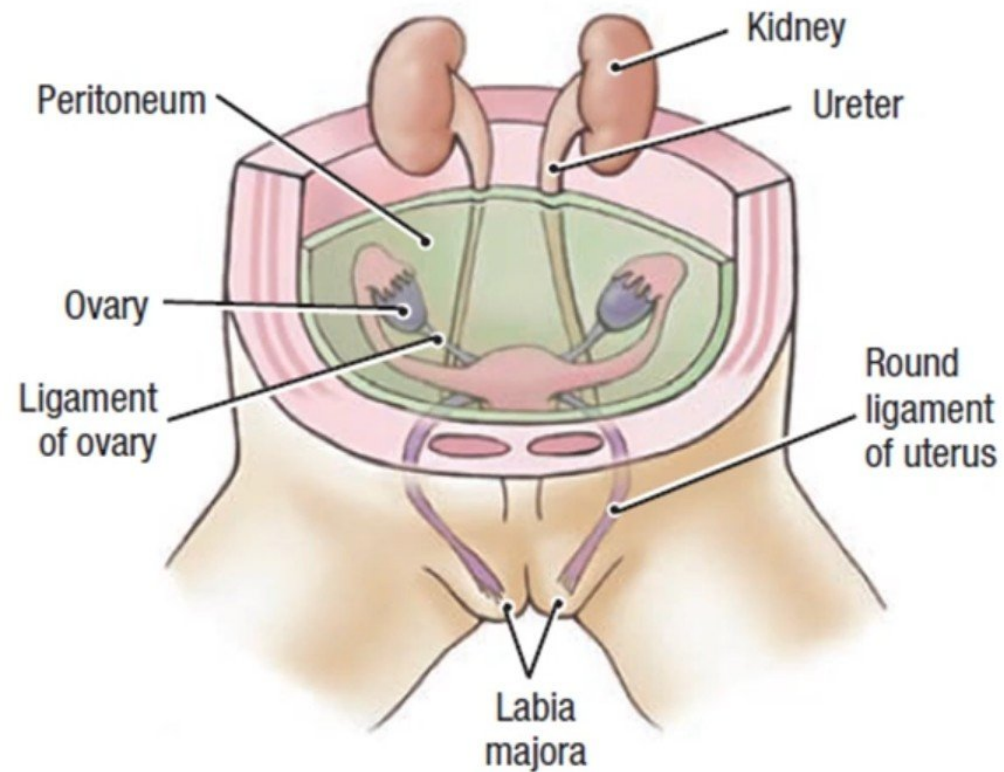
C. 9th month



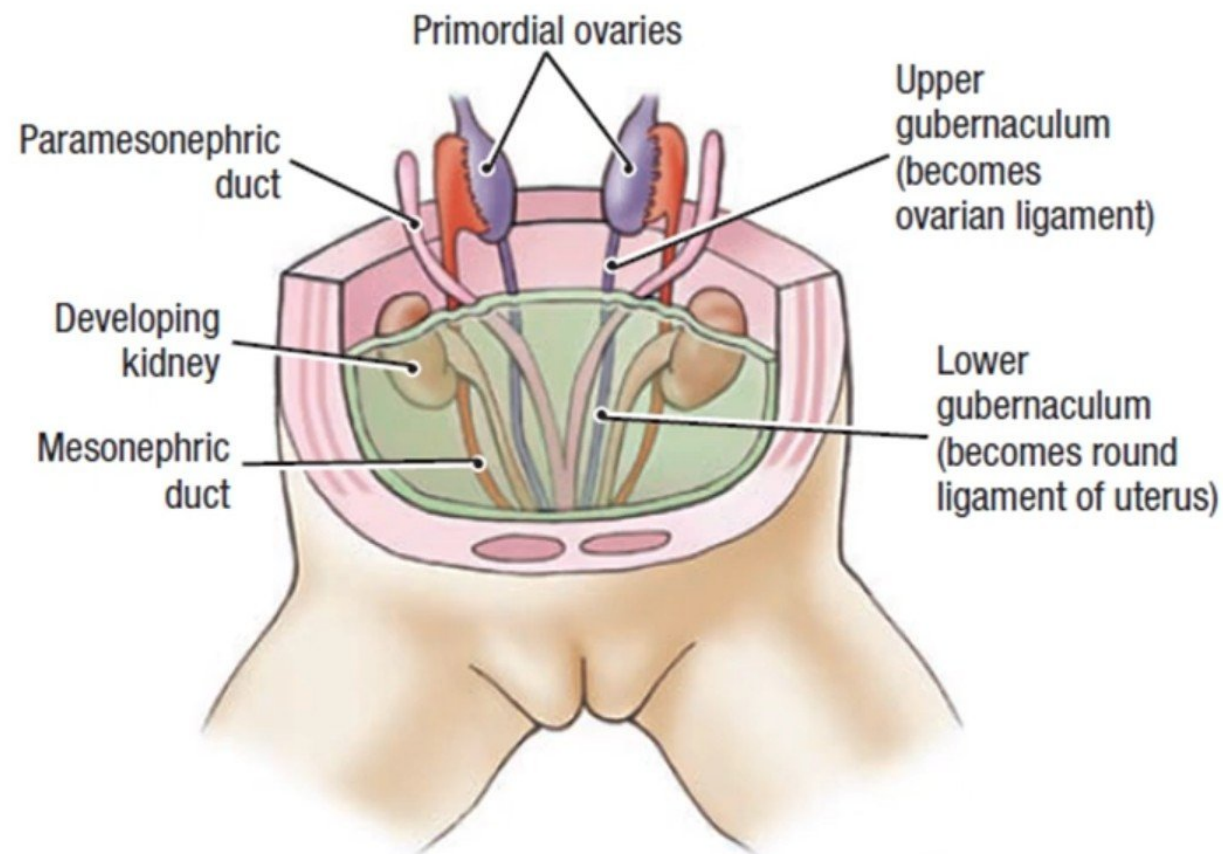
B. 7th month



Descent of gonads in female



Anterior View



Anterior View

Applied clinical anatomy

- A **hernia** is a **protrusion of tissue** (usually **parietal peritoneum** and **viscera** such as **fat, gut** or **omentum**) through or **alongside an opening in the abdomen** that is designed to allow a normal structure to enter or exit.

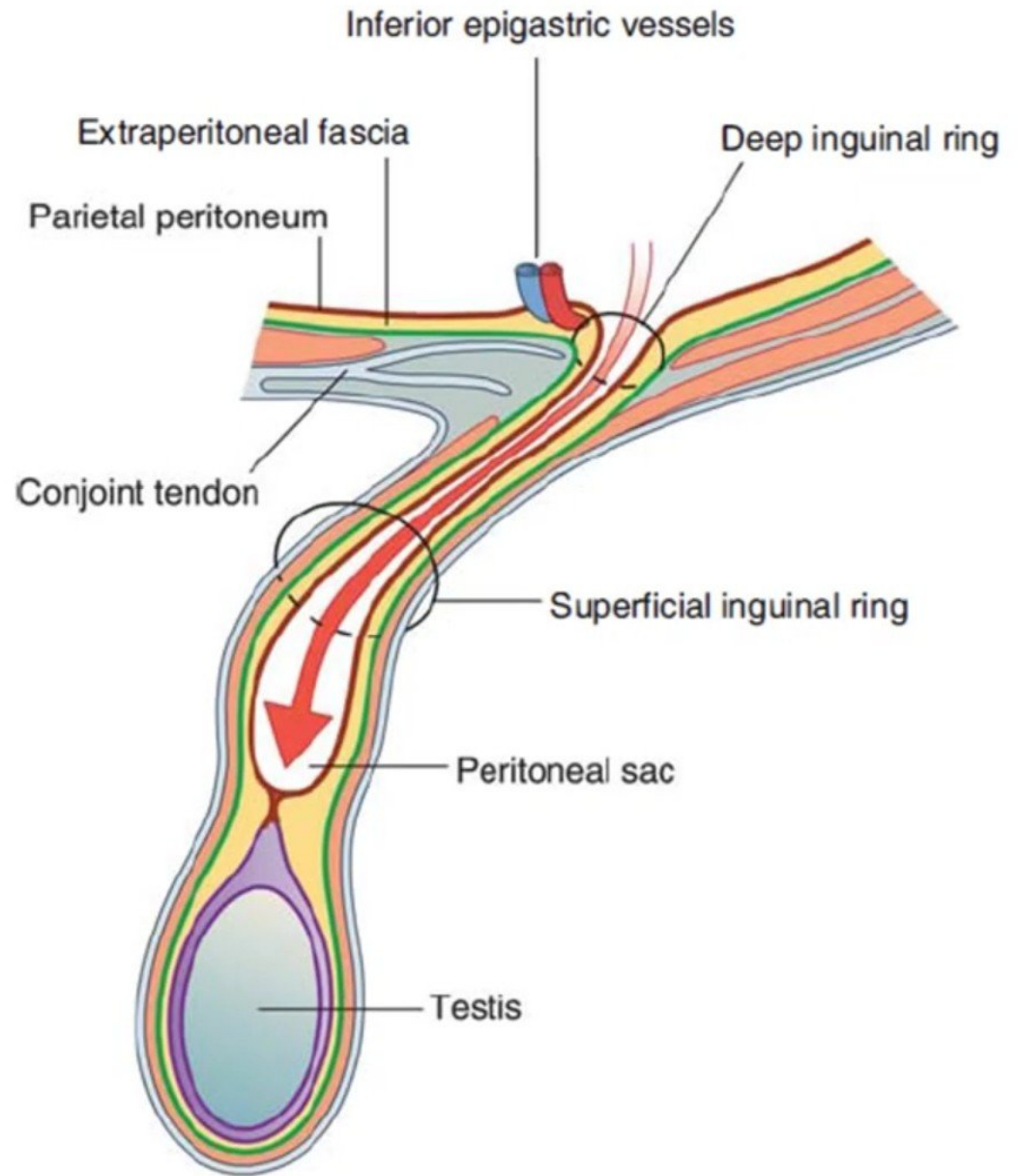
Example:

The **deep inguinal ring** may allow a hernia to appear **alongside the spermatic cord**.

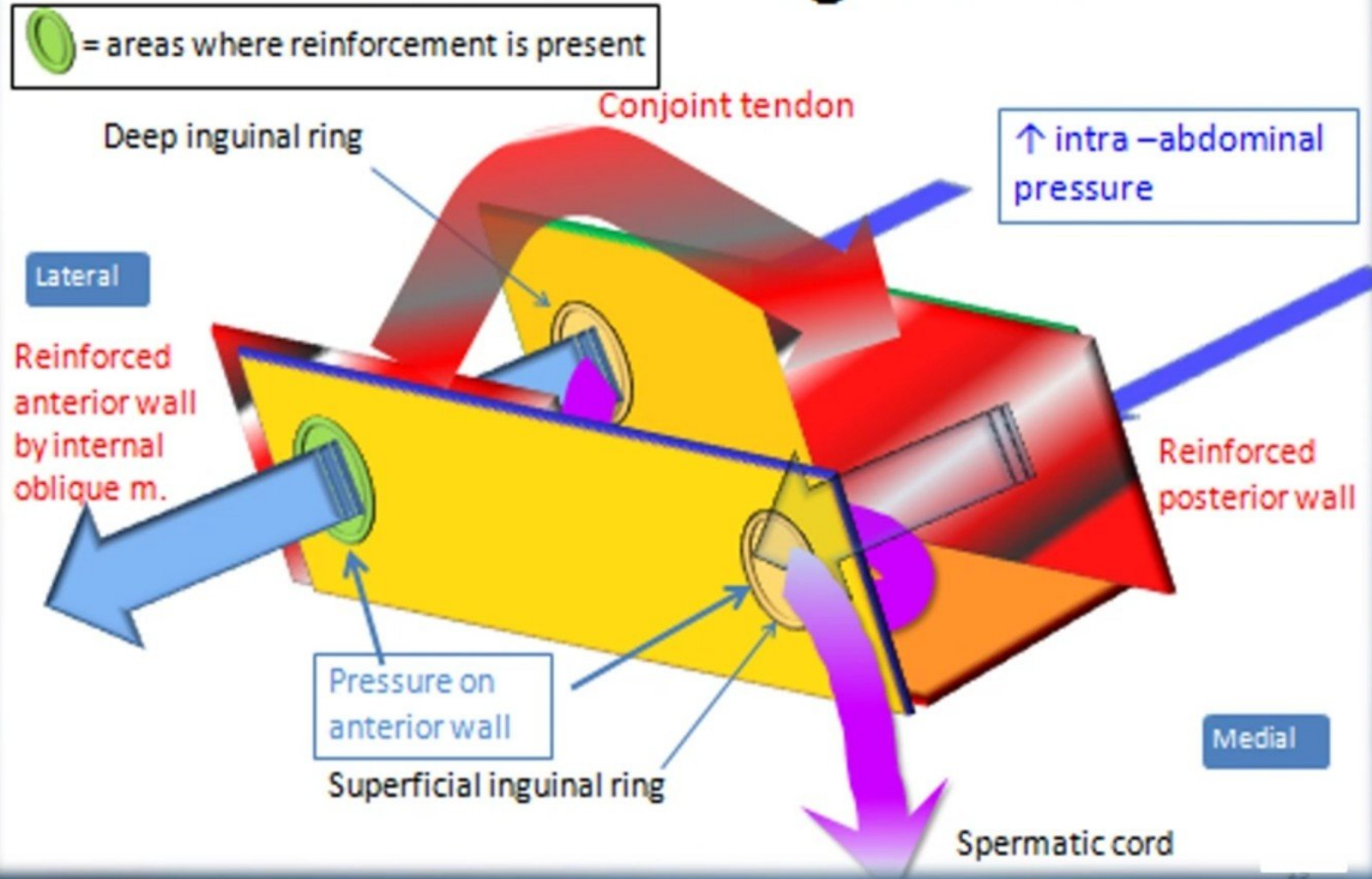
The **femoral canal** a hernia **alongside the lymphatics**.

Indirect (oblique hernias)

- This is the **most common of all abdominal hernias**.
- It leaves the abdominal cavity **lateral** to the **inferior epigastric vessels** and enters the **deep inguinal ring**.
- The hernial sac is formed by a persistent **processus vaginalis** and is **surrounded by all three fascial coverings** of the spermatic cord.
- The **hernia traverses** the entire **inguinal canal**.
- It **exits** through the **superficial inguinal ring** and commonly **enters the scrotum**.



Pressures on the inguinal canal



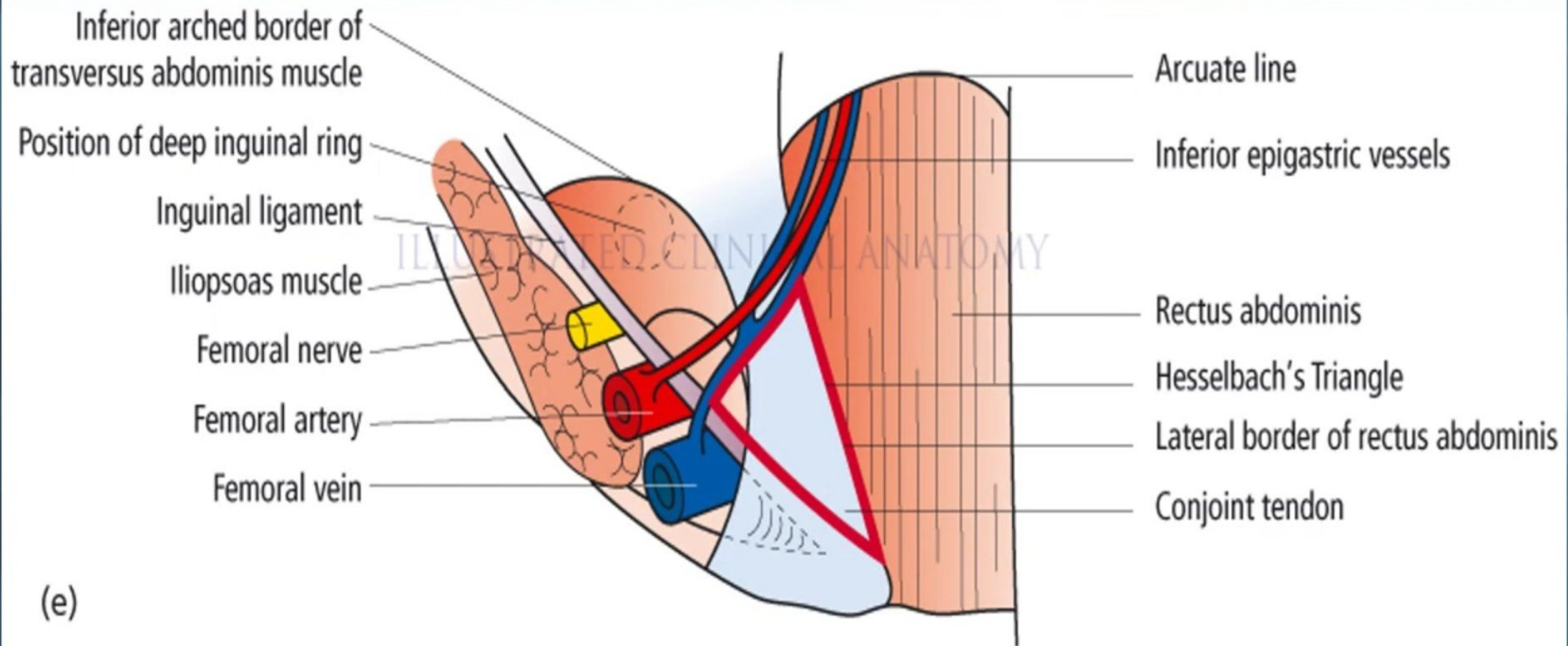
Coverings of indirect hernias

- 
- Peritoneum
 - Internal spermatic fascia (from transversalis fascia)
 - Cremaster muscle & fascia (from transversus abdominis and internal oblique mm.)
 - External spermatic fascia (from external oblique m.)
 - Superficial fascia
 - **Skin**

This is a list that you can reason out yourself. Work out the covering layers based on the abdominal wall layers.

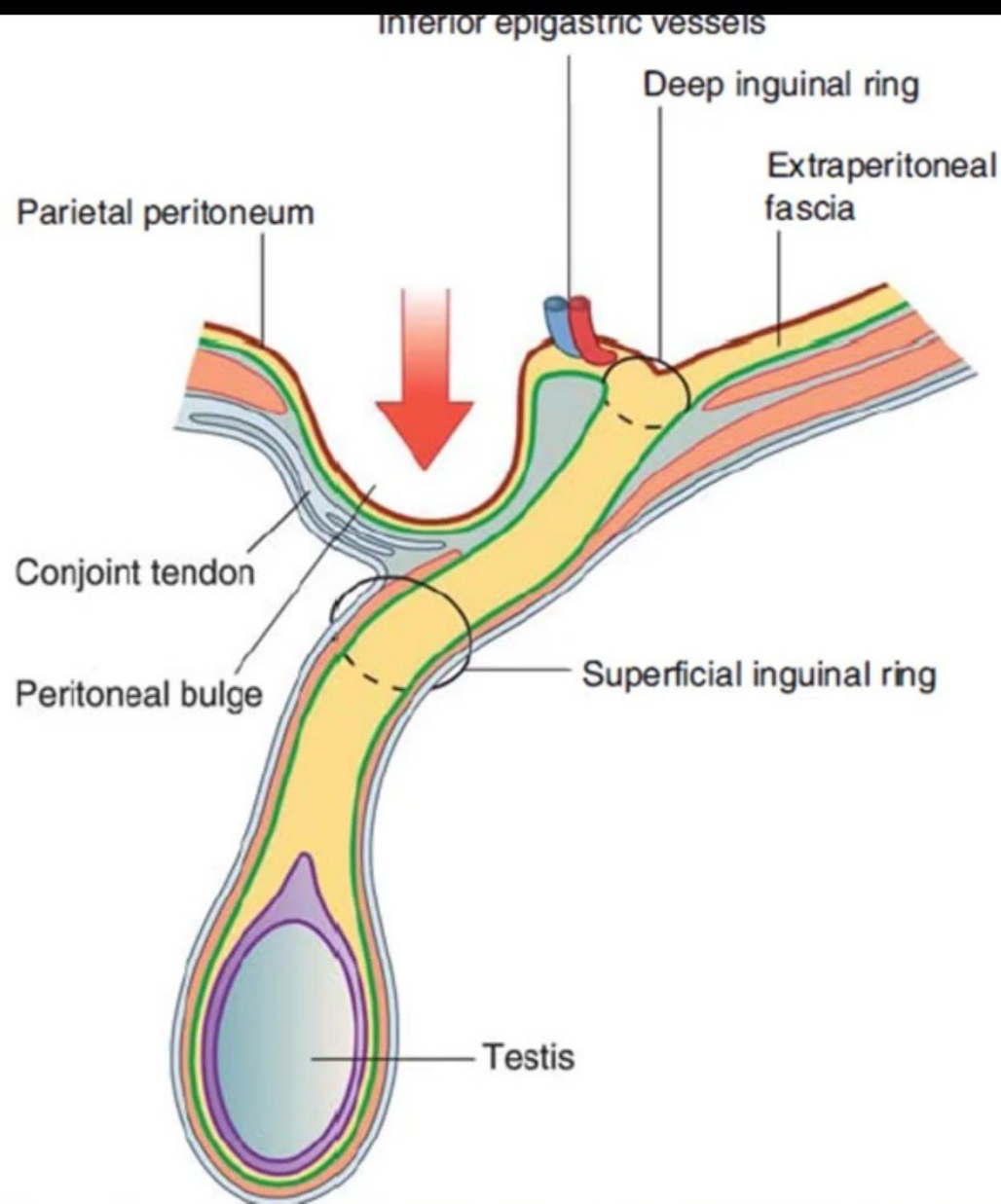


Inguinal triangle



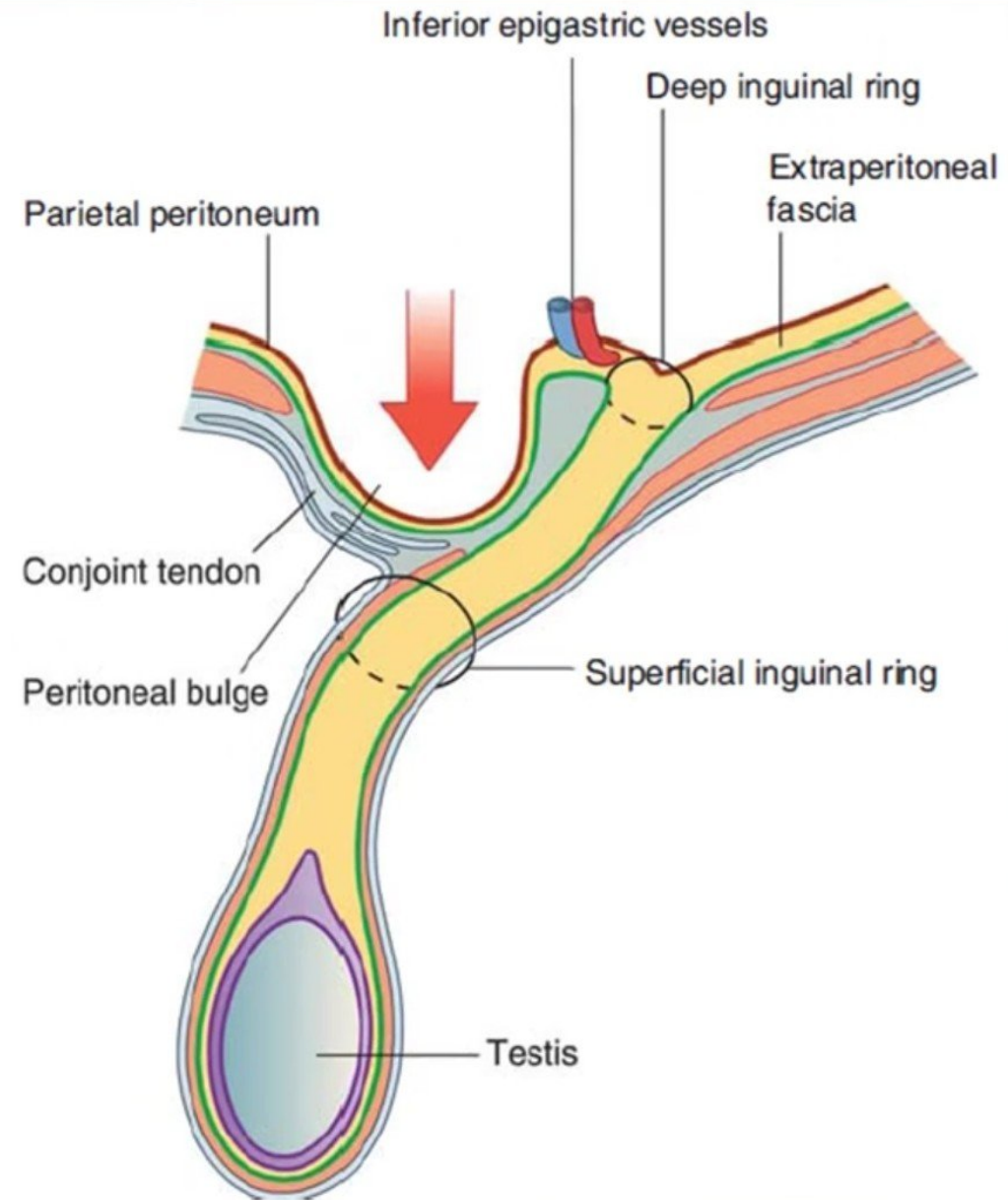
Direct hernias

- This is also known as **acquired inguinal hernia**.
- It is common in **elderly men**.
- The sac leaves the abdominal cavity **medial** to the **inferior epigastric artery**.
- It protrudes through an area of relative **weakness** in the **posterior wall of the inguinal canal**.
- The hernial sac is formed by **transversalis fascia**.
- It lies outside the **processus vaginalis**, which is usually obliterated, **parallel to the spermatic cord** and outside the inner one or **two fascial coverings of the cord**.

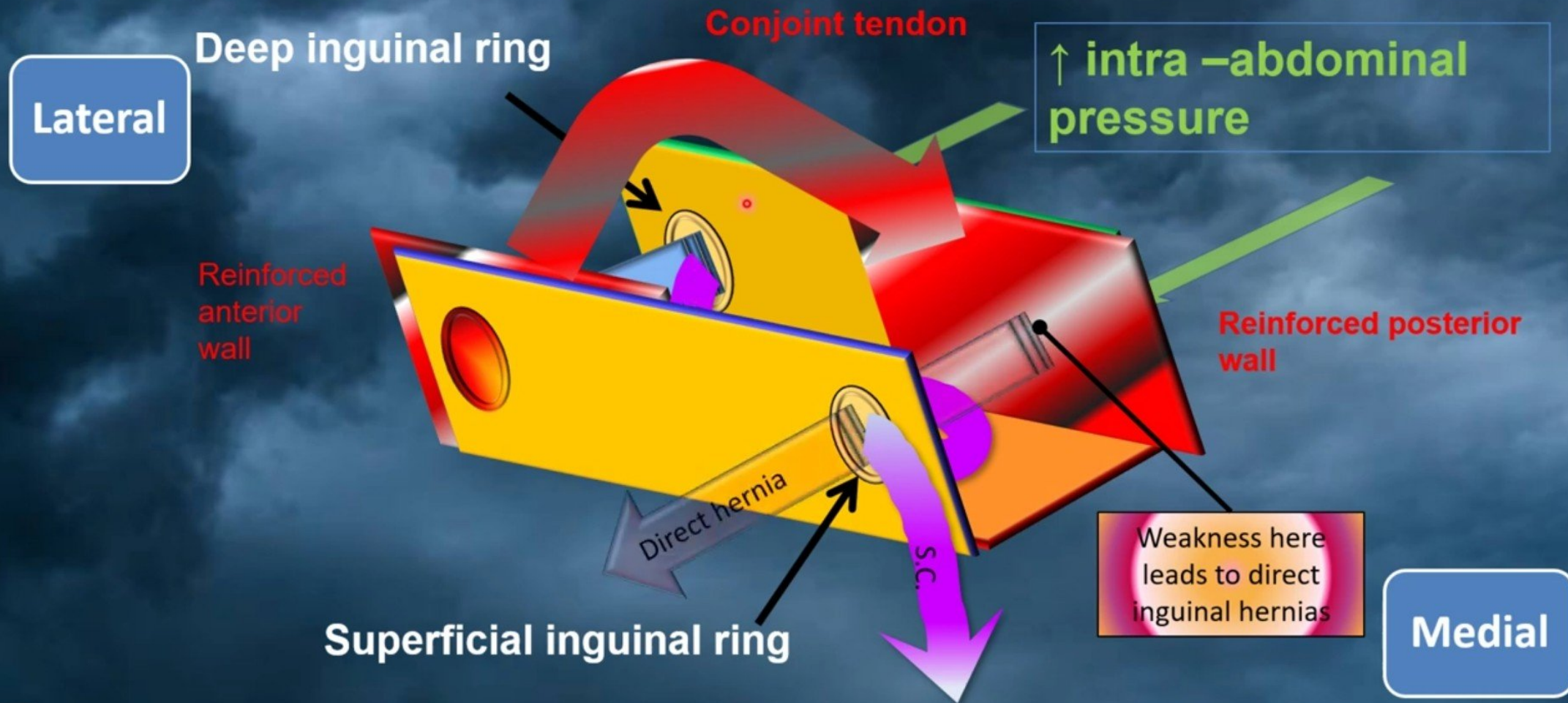


Direct hernias

- It does not entire the inguinal canal.
- The hernia protrudes through **the inguinal triangle** of Hesselbach that lies between **the inferior epigastric artery superolaterally**, **the rectus abdominis medially** and the **inguinal ligament inferiorly**.
- It emerges through or around the **conjoint tendon** to reach the **superficial inguinal ring**, gaining an outer **covering of external spermatic fascia** inside or parallel to that on the cord.
- It almost **never enters the scrotum**.



Pressures in the inguinal canal



Coverings of direct hernias

- 
- Peritoneum
 - Transversalis fascia
 - Conjoint tendon
 - External oblique aponeurosis
 - Superficial fascia
 - Skin

This is a list that you can reason out yourself. Work out the layers based on the anatomy. This will facilitate your understanding.



Inguinal triangle

The **inguinal triangle** (also known as **Hesselbach's triangle**) is an anatomical region of the lower abdominal wall, important in the context of **inguinal hernias**.

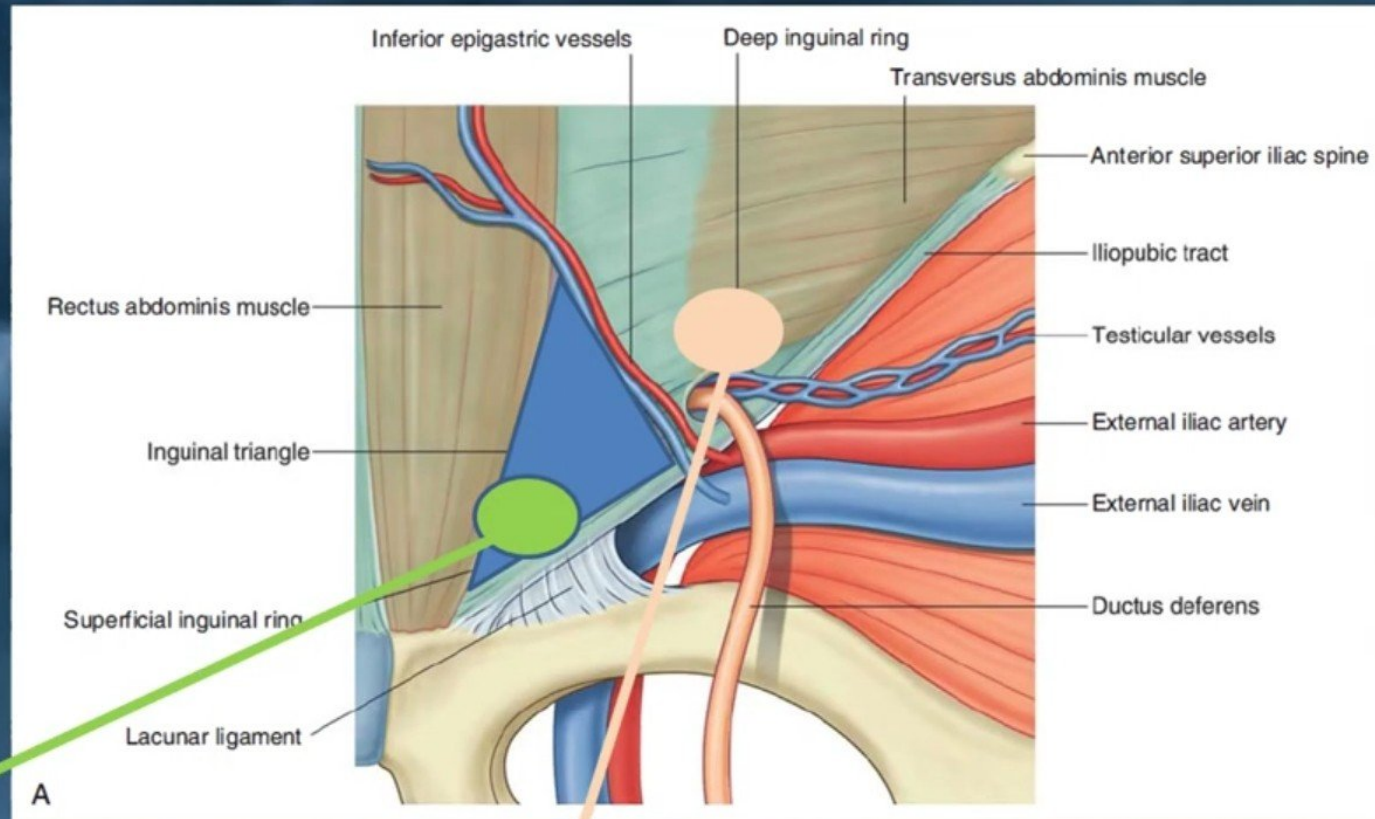
- It is a weak area where **direct inguinal hernias** can occur.

Borders of the Inguinal Triangle:

- 1. Medial Border** – Lateral edge of the rectus abdominis muscle.
- 2. Lateral Border** – Inferior epigastric vessels.
- 3. Inferior Border** – Inguinal ligament.

Clinical Significance:

- **Direct inguinal hernias** protrude through this triangle **medial** to the inferior epigastric vessels.



- **Indirect inguinal hernias** in contrast, pass **lateral** to the inferior epigastric vessels through the deep inguinal ring.

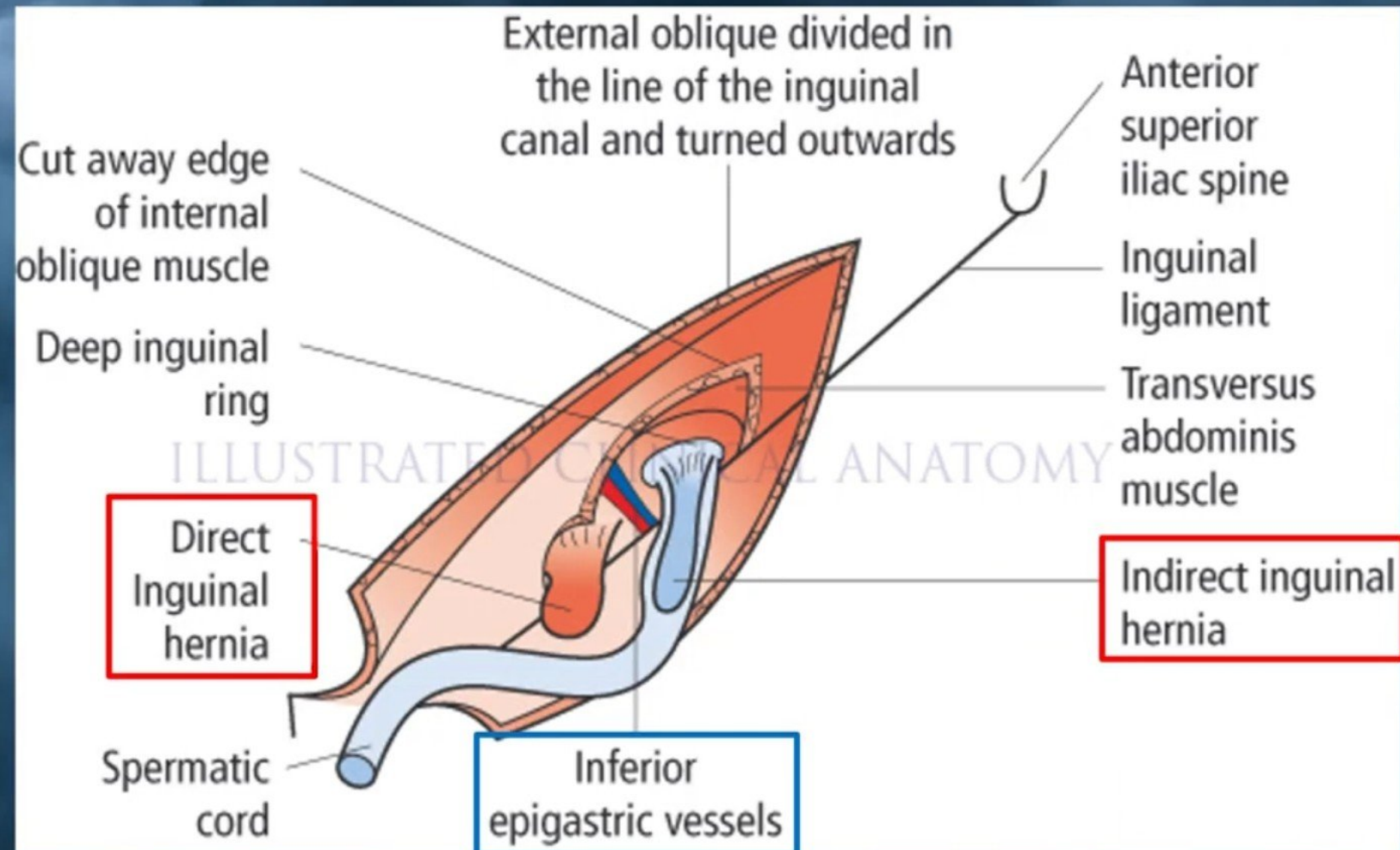
Relation of the types of hernias to inguinal triangle

Direct hernias:

- It arise **medial** to the **inferior epigastric vessels**.

Indirect hernias:

- It arise **lateral** to the **inferior epigastric vessels**.



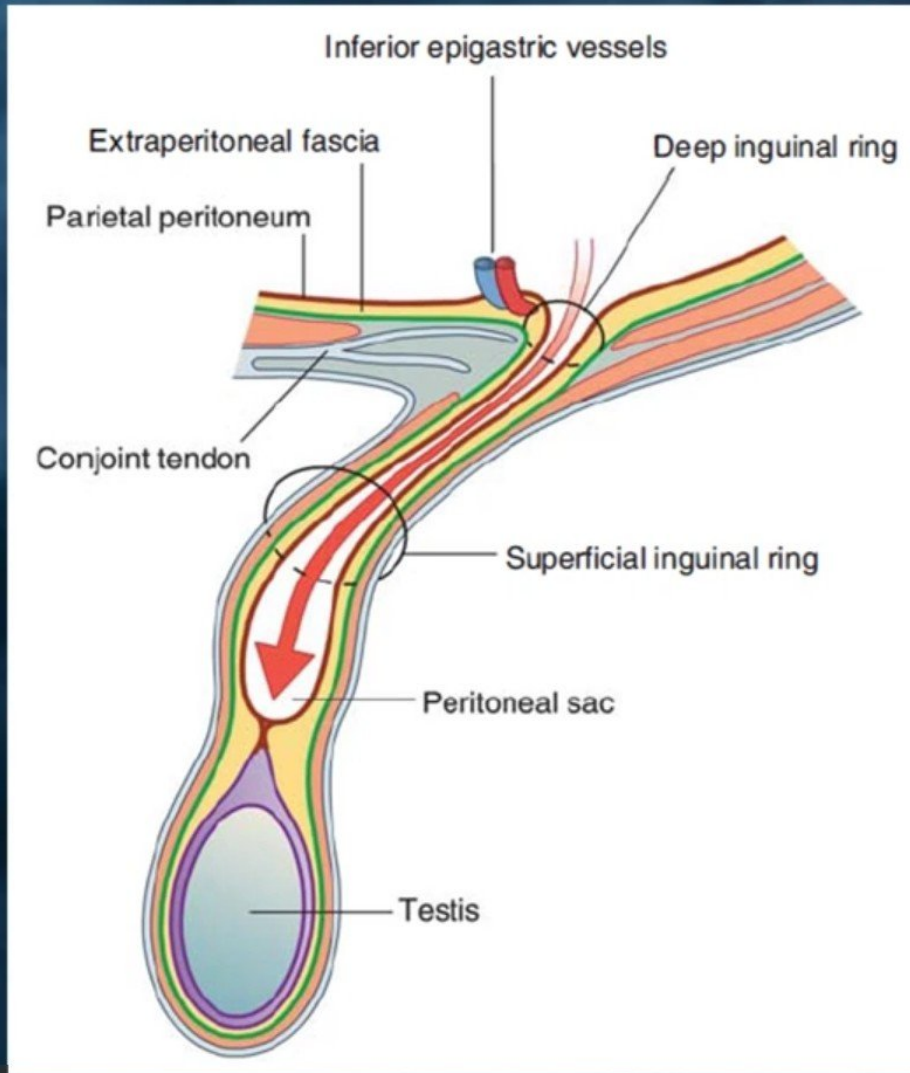
Summary of clinical correlation

- Inguinal hernia is a condition in which a portion of intestine protrudes through a weak spot in **the inguinal canal** or in **the inguinal triangle**.
- Inguinal hernia occurs **superior to the inguinal ligament** and **medial to the pubic tubercle**, occurs more in **males** than in **females**, and is derived from persistence of the **processus vaginalis**, which forms the tunica vaginalis
- Reducible hernia is a hernia in which **the contents of the hernial sac** can be **returned to** their **normal position**.
- Incarcerated hernia is an **irreducible hernia** in which the contents of **the hernial sac** are entrapped or stuck in the groin.
- Strangulated hernia is an irreducible hernia in which the intestine becomes tightly trapped or twisted; thus, the circulation is arrested, and gangrene (death of tissue) occurs unless relief is prompt. This is life threatening, and emergency surgical repair is required.

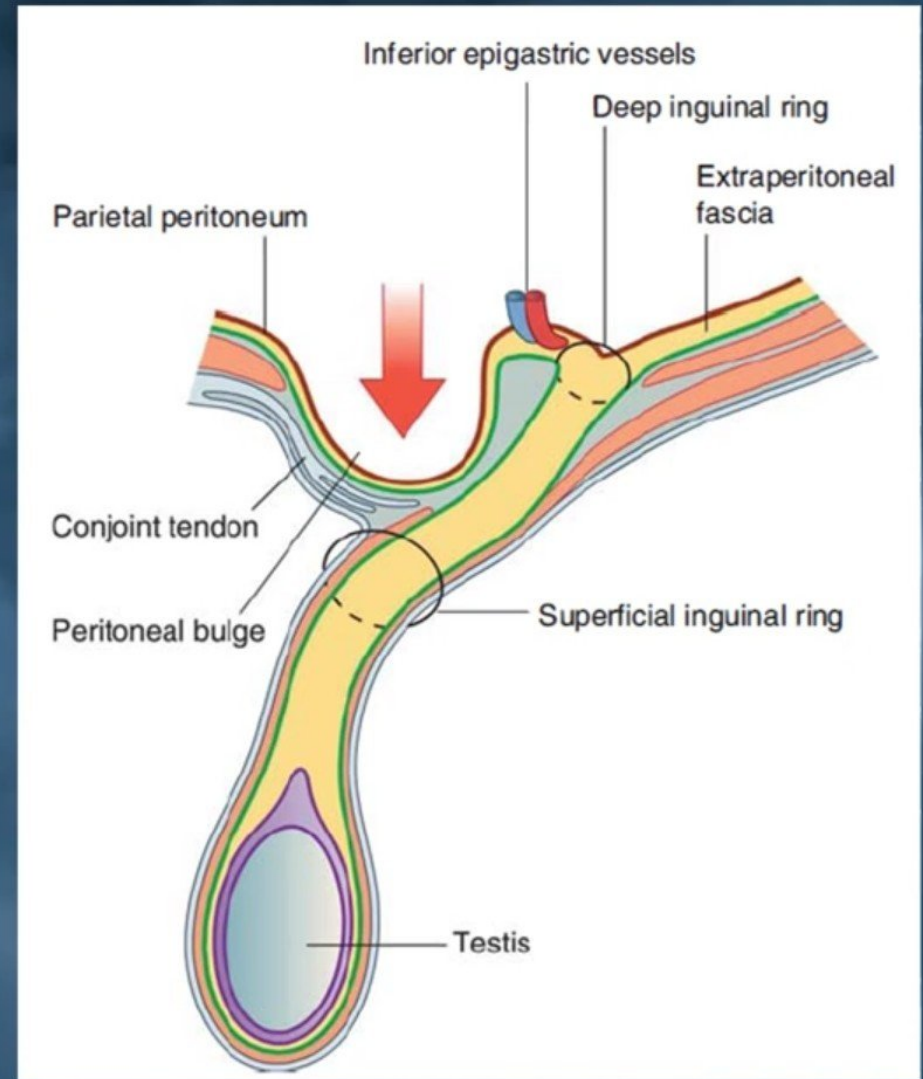
Summary of clinical correlation

- Indirect inguinal hernia passes through the deep inguinal ring, inguinal canal, and superficial inguinal ring and descends into the scrotum. The indirect hernia lies **lateral** to **the inferior epigastric vessels**, is **more common** than direct inguinal hernia, and is found more commonly on **the right side in men**.
- It is **congenital** (present at birth), associated with the persistence of the **processus vaginalis**, and covered by **the peritoneum** and **the coverings of the spermatic cord**.
- Direct inguinal hernia occurs **directly through a weakened area** of the abdominal wall muscles (**posterior wall of the inguinal canal**), lateral to the edge of the conjoint tendon (falx inguinalis), in the **inguinal triangle**, and **rarely descends into the scrotum**. The hernia lies **medial** to the **inferior epigastric vessels** and protrudes forward to (rarely through) the superficial inguinal ring.
- It is **acquired** (develops after birth) and has a sac formed by the **peritoneum** and occasionally the **transversalis fascia**.

Indirect inguinal hernia



Direct inguinal hernia



Left testicular venogram.

This image shows a **left testicular varicocele** before embolization.

Note: radiographs of varicoceles should be avoided to restrict radiation exposure.



Thanks